Algorithmic Trading Strategy

UMN FMA Trading Competition

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What We Built

Three autonomous strategies trading US equities & ETFs

Strategy	**Allocation**	**Asset(s)**
LSTM Portfolio	90%	VTI, SCHZ, PDBC, VIXM
RSI-2 Mean Reversion	5%	QQQ
Price vs 10-SMA	5%	SPY

All trades execute daily before market close via Alpaca API

Strategy 1: LSTM Portfolio (90%)

A neural network decides how to split money across 4 asset classes **every day**

- **VTI**: US Stocks
- **SCHZ**: Bonds
- **PDBC**: Commodities
- **VIXM**: Volatility

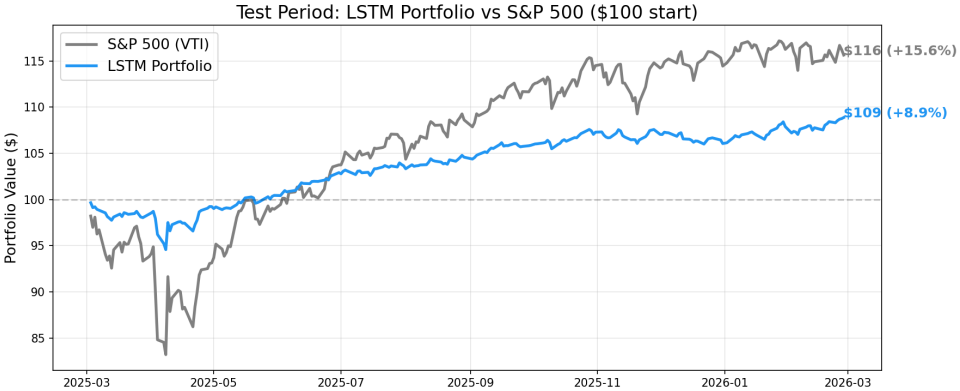
Trained on **10 years** of daily data (2016–2026)
Optimized for **Sharpe ratio**: maximizes returns *per unit of risk*

How the LSTM Works

1. Feeds in last **50 days** of prices & returns for all 4 assets
2. LSTM processes the sequence, learns patterns
3. Outputs **4 weights** (e.g., 28% stocks, 13% bonds, 30% commodities, 29% volatility)
4. Rebalances portfolio to match

Key: In uncertain markets, the model *automatically* shifts toward bonds, commodities, and volatility

LSTM Backtest Results



S&P 500: +15.6% return
Max drawdown: −16%

Our LSTM: +8.9% return
Max drawdown: **only −5%**

Why Lower Return is Actually Better Here

The competition scores **risk-adjusted** performance

- April 2025 crash: S&P dropped 16%, our LSTM only 5%
- **3× less drawdown** for roughly half the return
- In current uncertain market (tariffs, geopolitics), capital preservation matters

Strong returns with small losses > large returns with large losses

Strategy 2: RSI-2 Mean Reversion on QQQ (5%)

RSI (Relative Strength Index) measures how overbought or oversold an asset is

We use a 2-day RSI, computed from the last 2 daily price changes:

$$\mathrm{RSI}(2) = 100 - \frac{100}{1 + \dfrac{\text{average gain over 2 days}}{\text{average loss over 2 days}}}$$

- RSI near 0 = extremely oversold (price dropped sharply)
- RSI near 100 = extremely overbought (price surged)

Our rules:

- **Buy QQQ** when $\mathrm{RSI}(2) < 10$: sharp drop, expect a bounce
- **Sell QQQ** when $\mathrm{RSI}(2) > 70$: let profits run, then take profit
- Typical hold: 1–5 days

RSI-2 Backtest: Performance During Market Crises

Period	Dates	S&P 500	QQQ	RSI-2	RSI-2 Sharpe
Russia-Ukraine War	01-Feb-22 to 30-Jun-22	−16.1%	−22.8%	**+2.0%**	0.32
2022 Bear Market	01-Jan-22 to 31-Dec-22	−19.5%	−33.0%	**+3.3%**	0.26
2025 Tariff Crisis	01-Mar-25 to 30-May-25	−0.8%	+2.2%	**+13.4%**	**2.01**
Recent 3 Months	01-Dec-25 to 27-Feb-26	+0.4%	−1.9%	**+1.9%**	0.80
Full Period	27-Jul-20 to 27-Feb-26	+112.3%	+133.5%	**+140.9%**	**1.14**

Key insight: RSI-2 stayed positive in *every* crisis while S&P 500 lost 16–20%.
Max drawdown: only −10.6% vs −25.4% for S&P 500.

Strategy 3: Price vs 10-day SMA on SPY (5%)

SMA (Simple Moving Average) smooths out daily noise to reveal the trend

The 10-day SMA is the average closing price over the last 10 trading days:

$$\text{SMA}(10) = \frac{\text{Close}_1 + \text{Close}_2 + \cdots + \text{Close}_{10}}{10}$$

Our rules:

- **Buy SPY** when current price $>$ 10-day SMA (uptrend confirmed)
- **Sell SPY** when current price $<$ 10-day SMA (trend broken)
- Typical hold: 3–7 days

Complements RSI-2: RSI buys *fear* (dips), SMA rides *momentum* (trends)

Risk Management

1. **Diversified**: 4 asset classes, not just stocks
2. **AI-driven defense**: LSTM shifts to safe havens automatically
3. **Position sizing**: 90/5/5 split limits any single strategy's impact
4. **Long-only**: no leverage, no shorts, no options
5. **Daily rebalancing**: never holds stale positions
6. **All orders before close**: fills at real-time prices

Execution

Infrastructure

- Linux server running 24/7
- Python + PyTorch + Alpaca API
- 3 systemd services with auto-restart on failure

Daily Schedule (ET)

- 3:45 PM: LSTM fetches data, runs model, rebalances 4 ETFs
- 3:50 PM: RSI-2 checks QQQ signal, trades if triggered
- 3:50 PM: SMA-10 checks SPY signal, trades if triggered
- All orders fill before 4:00 PM close
- Max 4–6 orders per day

Summary

1. **LSTM (90%)**: AI allocates across stocks, bonds, commodities, volatility
2. **RSI-2 (5%)**: Buys QQQ dips, sells after recovery
3. **SMA-10 (5%)**: Rides SPY momentum, exits on breakdown

Designed for uncertain markets: protects capital while capturing upside

github.com/verma-bipul/Algo-Trade-UMN-Team4